

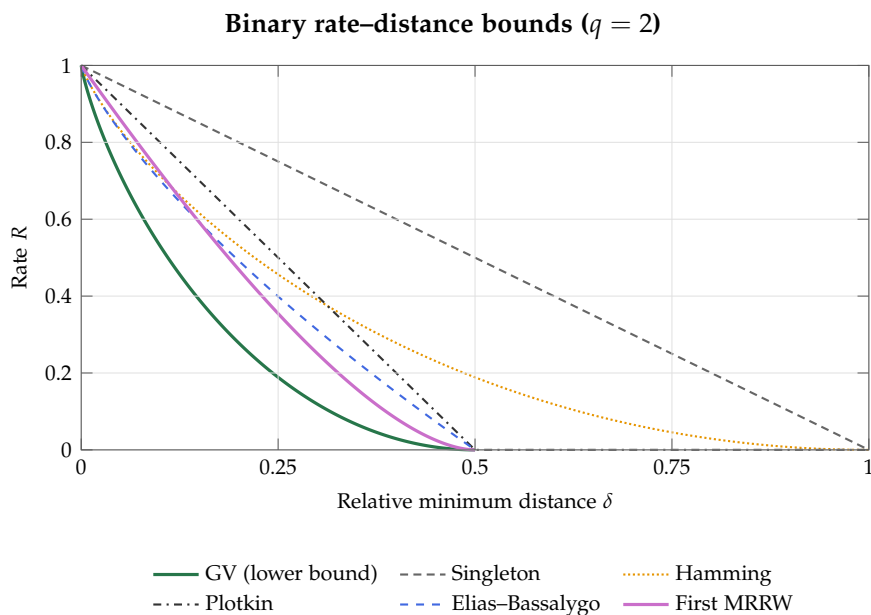
Topics in Error-Correcting Codes

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Lecture 4

Warning: This document is a rough draft, so it may contain bugs. Please feel free to email me with corrections.



GV is a lower bound; the other curves are upper bounds. The second MRRW bound and the 2026 improvements are not plotted.

Channel Decoding

Recall the communication model from Lecture 1: encode a message, send it through a channel, and decode the received word.

$$m \rightarrow \boxed{\text{Enc}} \rightarrow x \in \mathcal{C} \rightarrow \boxed{\text{channel ch}} \rightarrow y \rightarrow \boxed{\text{Dec}} \rightarrow \hat{m}$$

Theorem 1 (Distance and decoding; Lecture 1). *A code of minimum distance d has a decoder that corrects every pattern of $t < d/2$ errors. Equivalently, for every message m and every t -error channel ch ,*

$$\text{Dec}(\text{ch}(\text{Enc}(m))) = m.$$

This is the Hamming (adversarial) model of noise: a combinatorial guarantee for every message and *all* channels causing at most t errors. In practice, we may care

about random noise rather than an actual adversary. When can we decode then? Can we do better than $d/2$? These are the questions that motivate Shannon's model.

The binary symmetric channel

Definition 2 (BSC_p). For $0 \leq p \leq 1/2$, the *binary symmetric channel* BSC_p flips each transmitted bit independently with probability p . On a block $x \in \{0, 1\}^n$, its output is

$$Y = x \oplus Z, \quad Z_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(p).$$

Definition 3 (Capacity of a channel). Fix a memoryless channel W .¹ A rate R is *achievable* for W if there are encoders and decoders designed for this channel, of block lengths tending to infinity and asymptotic rate at least R , whose error probability when transmitting through W satisfies

$$P_{\text{err}}^{(n)} = \max_m \Pr[\text{Dec}(Y) \neq m \mid X = \text{Enc}(m)] \longrightarrow 0.$$

Here X, Y are the channel input and output. The *capacity* of W , denoted $\text{Cap}(W)$, is the supremum of rates achievable for this particular channel.

Theorem 4 (Capacity of BSC_p ; Shannon). For $0 \leq p \leq 1/2$,

$$\text{Cap}(\text{BSC}_p) = 1 - H_2(p), \quad H_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p).$$

Proof. Assume first that $0 < p < 1/2$.

Achievability. Fix $0 < R < 1 - H_2(p)$. By continuity, choose $p < \rho < 1/2$ such that $R < 1 - H_2(\rho)$. Choose a code uniformly among all M -element subsets of $\{0, 1\}^n$, where $M = \lfloor 2^{Rn} \rfloor$. Decode to the unique codeword within distance ρn of the received word, declaring failure if there is no such unique codeword.

When c is sent, an error can occur only if the noise has weight greater than ρn , or another codeword lies within distance ρn of $c \oplus Z$. Conditioned on c and Z , each other codeword is uniform among the $2^n - 1$ words different from c . A union bound therefore gives

$$\mathbb{E}_c[\text{average message error}] \leq \Pr[\text{wt}(Z) > \rho n] + \frac{(M - 1)\text{Vol}_2(n, \lfloor \rho n \rfloor)}{2^n - 1} =: \varepsilon_n.$$

The first term tends to zero by concentration of the number of bit flips around pn .

The second does too, since $\text{Vol}_2(n, \lfloor \rho n \rfloor) \leq 2^{H_2(\rho)n}$ and $R + H_2(\rho) < 1$.

Some code thus has average message error at most ε_n . Discard the messages whose error exceeds $2\varepsilon_n$; at least $M/2$ remain. Removing codewords cannot increase the error of the remaining messages under this decoding rule. The resulting code has maximum error at most $2\varepsilon_n \rightarrow 0$ and rate tending to R .

Converse. Fix $\gamma > 0$. A noise vector z of weight w has probability $p^w(1 - p)^{n-w}$, so

$$-\frac{1}{n} \log_2 \Pr[Z = z] = -\frac{w}{n} \log_2 p - \left(1 - \frac{w}{n}\right) \log_2 (1 - p).$$

Since $\text{wt}(Z)/n \rightarrow p$ in probability, with probability $1 - o(1)$ the noise lies in a set T_n in which every individual vector has probability at most $2^{-n(H_2(p) - \gamma)}$.

¹ Each channel use depends only on its current input. Conditioned on all inputs, the outputs are independent across uses.

Consider any code with M messages and maximum error tending to zero. Let D_m be the set of received words that its decoder maps to message m . These sets are disjoint, so $\sum_m |D_m| \leq 2^n$. For a uniformly chosen message, successful decoding with noise in T_n has probability $1 - o(1)$. For each $y \in D_m$, the only noise vector producing y from $\text{Enc}(m)$ is $y \oplus \text{Enc}(m)$. Consequently,

$$1 - o(1) \leq \frac{1}{M} \sum_m |D_m| 2^{-n(H_2(p) - \gamma)} \leq \frac{2^{n(1 - H_2(p) + \gamma)}}{M}.$$

Taking logarithms and dividing by n gives $R \leq 1 - H_2(p) + \gamma$. Letting $\gamma \rightarrow 0$ proves the upper bound.

Finally, $p = 0$ is noiseless and has capacity 1. For $p = 1/2$, the output is independent of the input, so the average success probability for M equiprobable messages is at most $1/M$; the capacity is 0. $\square$

The *binary erasure channel* BEC_p instead independently replaces each bit by an erasure symbol $?$ with probability p , revealing the locations of the missing bits. We will not study it today.

The Hamming–Shannon Gap and List Decoding

Fix $0 < p < 1/2$. Compare two requirements for recovering the message:

- **Shannon: random errors.** On BSC_p , decoding must succeed with probability tending to 1. Every rate $R < 1 - H_2(p)$ is achievable.
- **Hamming: arbitrary errors.** Decoding must succeed for every message and every pattern of at most $\lfloor pn \rfloor$ errors. Sphere packing gives $R \leq 1 - H_2(p)$.

Does the stronger Hamming requirement force a smaller rate? The Hamming–Shannon gap is the difference between the best achievable rates in these two models. Hamming’s bound alone does not show a gap: it matches the BSC capacity, but it is only an upper bound.

For $0 < p < 1/4$, GV applied at relative distance $2p$ gives Hamming-model rates approaching $1 - H_2(2p)$. We will use Elias–Bassalygo to prove that the best Hamming-model rate is strictly below $1 - H_2(p)$. For $1/4 \leq p < 1/2$, Plotkin already shows a gap: the Hamming-model rate is zero, while the BSC capacity is positive.

Allowing a short list

Does allowing a small list always buy us a decoding radius beyond $d/2$?

List decoding aims to go beyond $d/2$ errors by allowing a small list of candidate codewords containing the transmitted one. If the receiver has other context about the message, it may use that context to select the intended message from the list.

For a formal treatment using recognizable valid messages, see [Grossman and Holmgren \[2021\]](#); also see [Oded’s comments](#). I thank [Surendra Ghentiyala](#) for introducing me to this paper.

Definition 5 (List decoding). For $y \in \Sigma^n$ and $\rho \geq 0$, let

$$B_q(y, \rho n) = \{x \in \Sigma^n : \Delta(x, y) \leq \rho n\}, \quad \mathcal{L}(y, \rho) = \mathcal{C} \cap B_q(y, \rho n).$$

The code $\mathcal{C}$ is (ρ, L) -list-decodable if $|\mathcal{L}(y, \rho)| \leq L$ for every received word y . A list decoder outputs all the codewords in $\mathcal{L}(y, \rho)$.

Unique decoding is the case $L = 1$. This definition is combinatorial; efficient algorithms for finding the list are a separate question.

Theorem 6 (Binary list-decoding capacity). Fix $0 \leq \rho < 1/2$.

1. For every $\eta > 0$ with $\eta < 1 - H_2(\rho)$, there are binary codes of asymptotic rate at least $1 - H_2(\rho) - \eta$ that are (ρ, L) -list-decodable for $L = O(1/\eta)$, independent of n .
2. Any family of (ρ, L_n) -list-decodable binary codes with $\log_2 L_n = o(n)$ has rate at most $1 - H_2(\rho)$.

Proof. Write $N = 2^n$ and $V = \text{Vol}_2(n, \lfloor \rho n \rfloor)$ for the size of a Hamming ball. Recall that

$$V \leq 2^{H_2(\rho)n}, \quad \frac{\log_2 V}{n} = H_2(\rho) + o(1).$$

Upper bound. Count pairs (c, y) with $c \in \mathcal{C}$ and $\Delta(c, y) \leq \rho n$. Each codeword has exactly V possible centers, while each center has at most L_n codewords in its ball. Thus $|\mathcal{C}|V \leq NL_n$, and

$$\frac{\log_2 |\mathcal{C}|}{n} \leq 1 - \frac{\log_2 V}{n} + \frac{\log_2 L_n}{n} = 1 - H_2(\rho) + o(1),$$

where the last step uses $\log_2 L_n = o(n)$.

Existence. Set $R = 1 - H_2(\rho) - \eta$, $M = \lfloor 2^{Rn} \rfloor$, and $L = \lceil 2/\eta \rceil$. For sufficiently large n , choose $\mathcal{C}$ uniformly among all M -element subsets of $\{0, 1\}^n$. Fix a center y and a set S of $L + 1$ distinct words in $B_2(y, \rho n)$. Sampling without replacement gives

$$\Pr[S \subseteq \mathcal{C}] = \prod_{j=0}^L \frac{M-j}{N-j} \leq \left(\frac{M}{N}\right)^{L+1}.$$

The code fails to be (ρ, L) -list-decodable exactly when some such pair (y, S) has $S \subseteq \mathcal{C}$. There are $N \binom{V}{L+1}$ possible pairs, so a union bound gives

$$\begin{aligned} \Pr[\text{failure}] &\leq N \binom{V}{L+1} \left(\frac{M}{N}\right)^{L+1} \\ &\leq N \left(\frac{MV}{N}\right)^{L+1} \leq 2^{n(1-\eta(L+1))} \leq 2^{-n}. \end{aligned}$$

Here $MV/N \leq 2^{-\eta n}$ since $R + H_2(\rho) = 1 - \eta$. Hence a suitable code exists for every sufficiently large n . Its rate $\log_2 M/n$ tends to R , and its list size is $L = O(1/\eta)$, independent of n . $\square$

List decoding therefore reaches the Shannon rate $1 - H_2(p)$ against an adversarial fraction p of errors, up to any fixed rate slack. It does so by relaxing the output to a list; it does not supply a unique answer at that rate.

The random construction shows what is possible for some codes. Our initial question asks whether list decoding improves on unique decoding for *every* code. The Johnson bound will answer this using only the code's distance.

Remark (Generalized Singleton bound). For fixed list size L , a family of positive asymptotic rate R that is (ρ, L) -list-decodable satisfies

$$\rho \leq \frac{L}{L+1}(1-R), \quad \text{equivalently} \quad R \leq 1 - \frac{L+1}{L}\rho.$$

Finite-length versions include rounding and a term $\log_q L/n$. See [Shangguan and Tamo \[2023\]](#) and [Roth \[2022\]](#). For $L = 1$, this is the usual Singleton restriction $\rho \leq (1-R)/2$ on unique decoding. As L grows, the bound approaches $\rho \leq 1-R$, the large-alphabet benchmark.

Elias's improvement: a preview

For a family of binary codes of relative distance at least $\delta \in (0, 1/2)$, the *Elias–Bassalygo bound* says

$$R \leq 1 - H_2\left(\frac{1 - \sqrt{1 - 2\delta}}{2}\right). \quad (1)$$

We will prove this using list decoding. The idea is to show that a ball of radius larger than $\delta n/2$ still contains only a small number of codewords, and then count how many such balls cover the code. The needed local statement is the Johnson bound.

The Johnson Bound

Fix a received word y and its list of L codewords. Let A count agreements over all coordinates and *ordered* pairs of list codewords, including a codeword paired with itself. Minimum distance δn gives

$$A \leq Ln + L(L-1)(1-\delta)n. \quad (2)$$

We will lower-bound A using agreements with the common center y .

An alphabet-independent bound

Theorem 7 (Johnson bound, independent of q). *Let $\mathcal{C} \subseteq \Sigma^n$ have relative distance at least $0 < \delta \leq 1$. Define*

$$J(\delta) = 1 - \sqrt{1 - \delta}.$$

For every $0 \leq \rho < J(\delta)$, the code is $(\rho, (J(\delta) - \rho)^{-2})$ -list-decodable, regardless of the alphabet size.

Proof. Assume $L > 0$, and let a_i count the list codewords that agree with y_i . Each codeword agrees with y in at least $(1 - \rho)n$ coordinates, so $\sum_i a_i \geq L(1 - \rho)n$. These codewords contribute a_i^2 ordered agreements at coordinate i . By Cauchy–Schwarz,

$$A \geq \sum_{i=1}^n a_i^2 \geq \frac{(\sum_i a_i)^2}{n} \geq L^2(1 - \rho)^2n.$$

Comparing with (2) and rearranging gives

$$L(\delta - 2\rho + \rho^2) \leq \delta.$$

Writing $\eta = J(\delta) - \rho > 0$, the coefficient is $2\eta\sqrt{1 - \delta} + \eta^2 \geq \eta^2$. Since $\delta \leq 1$, we get $L \leq 1/\eta^2$. $\square$

For $0 < \delta \leq 1$, rationalizing shows that

$$J(\delta) = \frac{\delta}{1 + \sqrt{1 - \delta}} > \frac{\delta}{2}.$$

This answers our initial question affirmatively: for every code of fixed relative distance $\delta > 0$, any radius strictly between $\delta/2$ and $J(\delta)$ admits a list size independent of n . Thus every such code gains a larger asymptotic decoding radius, not only the random codes from the capacity construction. For Reed–Solomon codes, $\delta \approx 1 - R$, so the radius approaches $1 - \sqrt{R}$ from below.

The binary improvement

Theorem 8 (Binary Johnson bound). *Let $\mathcal{C} \subseteq \{0, 1\}^n$ have relative distance at least $0 < \delta \leq 1/2$, and define*

$$J_2(\delta) = \frac{1 - \sqrt{1 - 2\delta}}{2}.$$

For every $0 \leq \rho < J_2(\delta)$, the code is $(\rho, (J_2(\delta) - \rho)^{-2})$ -list-decodable.

As in the Plotkin bound, knowing the alphabet gives more information about agreements: in the binary case, codewords that disagree with y_i must agree with one another. Counting these additional agreements improves the argument above. The proof is on Problem Set 1.

The Elias–Bassalygo Bound

Theorem 9 (Elias–Bassalygo bound). For fixed $q \geq 2$, put $\alpha = 1 - 1/q$ and $J_q(\delta) = \alpha(1 - \sqrt{1 - \delta/\alpha})$. For $0 < \delta < \alpha$, any family of q -ary codes with relative distance at least δ satisfies

$$R \leq 1 - H_q(J_q(\delta)).$$

In particular, the binary bound is $R \leq 1 - H_2((1 - \sqrt{1 - 2\delta})/2)$.

Proof. Fix $\rho < J_q(\delta)$, and consider a code whose minimum distance is at least δn . For a uniformly random center $Y \in \Sigma^n$, double counting gives

$$\mathbb{E}_Y[|\mathcal{C} \cap B_q(Y, \rho n)|] = \frac{|\mathcal{C}| \text{Vol}_q(n, \lfloor \rho n \rfloor)}{q^n}.$$

By Johnson, $\mathcal{C}$ is (ρ, L) -list-decodable for a constant L . Thus

$$|\mathcal{C}| \leq \frac{Lq^n}{\text{Vol}_q(n, \lfloor \rho n \rfloor)}, \implies \frac{\log_q |\mathcal{C}|}{n} \leq 1 - H_q(\rho) + o(1).$$

As $n \rightarrow \infty$, we get $R \leq 1 - H_q(\rho)$ for every $\rho < J_q(\delta)$. Letting $\rho \uparrow J_q(\delta)$ proves the result. $\square$

We can now answer the Hamming–Shannon question. For $0 < p < 1/4$, a binary code family correcting every pattern of at most pn errors needs relative distance at least $2p$ asymptotically. Its rate satisfies

$$R \leq 1 - H_2(J_2(2p)) < 1 - H_2(p) = \text{Cap}(\text{BSC}_p).$$

The strict inequality follows from $J_2(2p) > p$ and the strict monotonicity of H_2 on $[0, 1/2]$. Thus correcting arbitrary errors requires a strictly lower rate than reliable decoding on BSC_p .

MRRW and Recent Improvements

Theorem 10 (First MRRW bound, 1977). For a 2-ary code family with asymptotic rate R and relative distance $0 \leq \delta \leq 1/2$,

$$R \leq H_2\left(\frac{1}{2} - \sqrt{\delta(1 - \delta)}\right).$$

Due to Robert J. McEliece, Eugene R. Rodemich, Howard Rumsey, Jr., and Lloyd R. Welch [McEliece et al., 1977].

For the full proof, see the reference textbook *Essential Coding Theory*, Section 8.2.

Recent improvements. The MRRW bounds were improved by OpenAI [2026] and Alrabiah and Guruswami [2026].

The Regime $\delta = 1/2 - \varepsilon$

Bounds on the rate R of binary code families as $\varepsilon \rightarrow 0^+$, taking $n \rightarrow \infty$ first. The upper bounds are ordered from weaker to stronger.

Bound	Type	Asymptotic value
Hamming	Upper	$1 - H_2(1/4) + O(\varepsilon)$
Plotkin	Upper	2ε
Elias–Bassalygo	Upper	$\varepsilon / \ln 2 + O(\varepsilon^2)$
First MRRW	Upper	$2\varepsilon^2 \log_2(1/\varepsilon) + O(\varepsilon^2)$
GV	Lower (achievable)	$2\varepsilon^2 / \ln 2 + O(\varepsilon^4)$

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